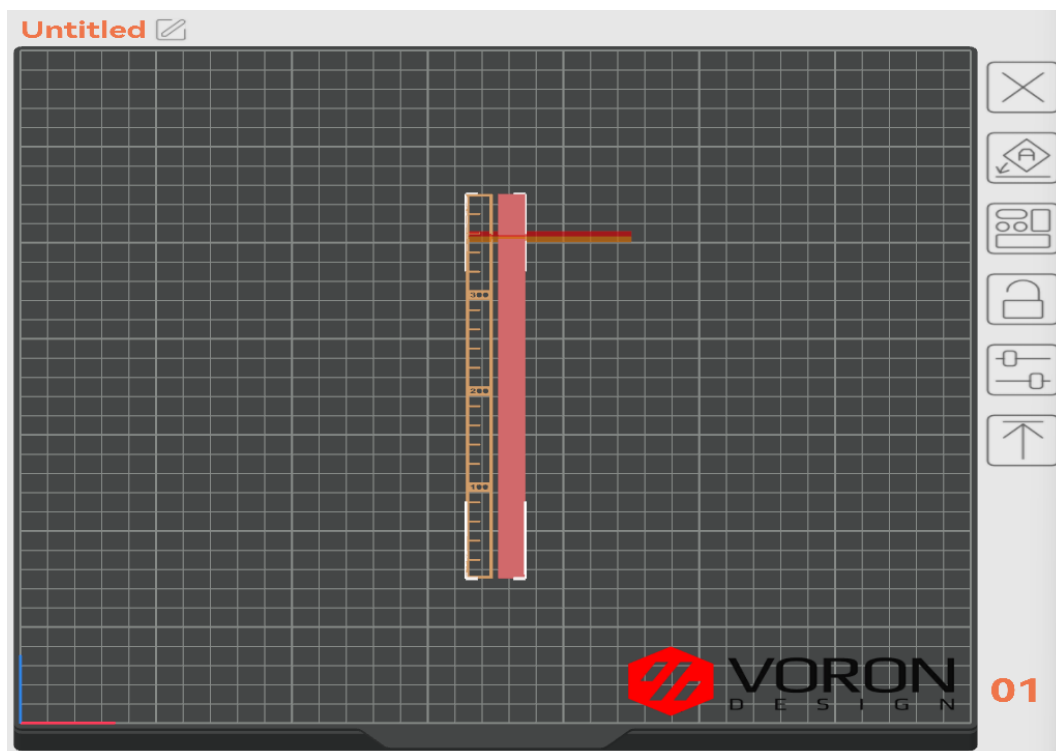


Using the Purge tuning and setup STL to determine flushing volumes

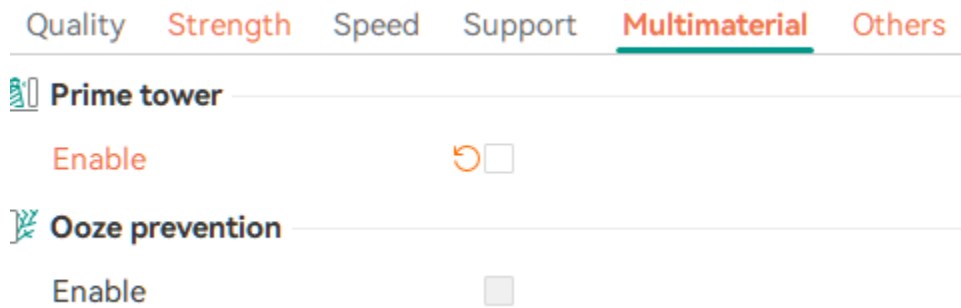
The print needed for this process is on Makerworld which is owned by Bambu. You will need to create an account to download it. It's free and easy to set up. Makerworld.com is the URL to the site.

I'm making this document because there are no instructions on how to use it that I could find. I have a Voron v2 350mm. Obviously your mileage may vary based on your printer. I also use Orca slicer so your slicer's setting screens might look different.

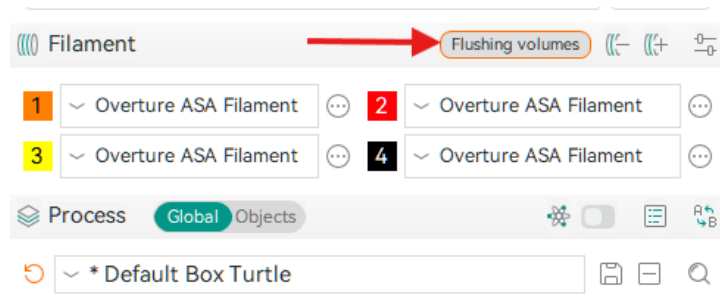
1. Download the .3mf file for the purge tuning print. It's a very small, single layer print that does a good job of indicating the setting needed moving from one color to another. The file is located here: <https://makerworld.com/en/models/510316-quick-easy-flushing-volume-scale-0-400#profileId-429735>
2. In the console window of Klipper, enter the command below to disable pooping as that will skew the results beyond use.
`AFC_TOGGLE_MACRO POOP=0`
3. In Orca Slicer, load the .3mf, and you will be presented with this build plate.



4. In Orca, go to the Multimaterial section of the settings, and unselect Purge Tower.



5. I have a 4 lane Box Turtle so I have 4 colors selected. The process is the same no matter the number of colors. You now need to add “objects” based on the number of color swaps you can do. To find this number, you can use the Flusing Volumes dialog in Orca.



6. Using the Flushing Volumes dialog, you will set the multiplier at the bottom of the dialog (indicated by the bottom arrow) to a value of 1. Then set the editable values in the from/to table all to zero. In my case there are 12 editable values indicating I need to set up for 12 different swaps.

Flushing volumes for filament change

Orca would re-calculate your flushing volumes everytime the filaments color changed or filaments changed. You could disable the auto-calculate in Orca Slicer > Preferences

Re-calculate

from/to	1	2	3	4
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0

Flushing volume (mm³) for each filament pair.
Suggestion: Flushing Volume in range [0, 900]

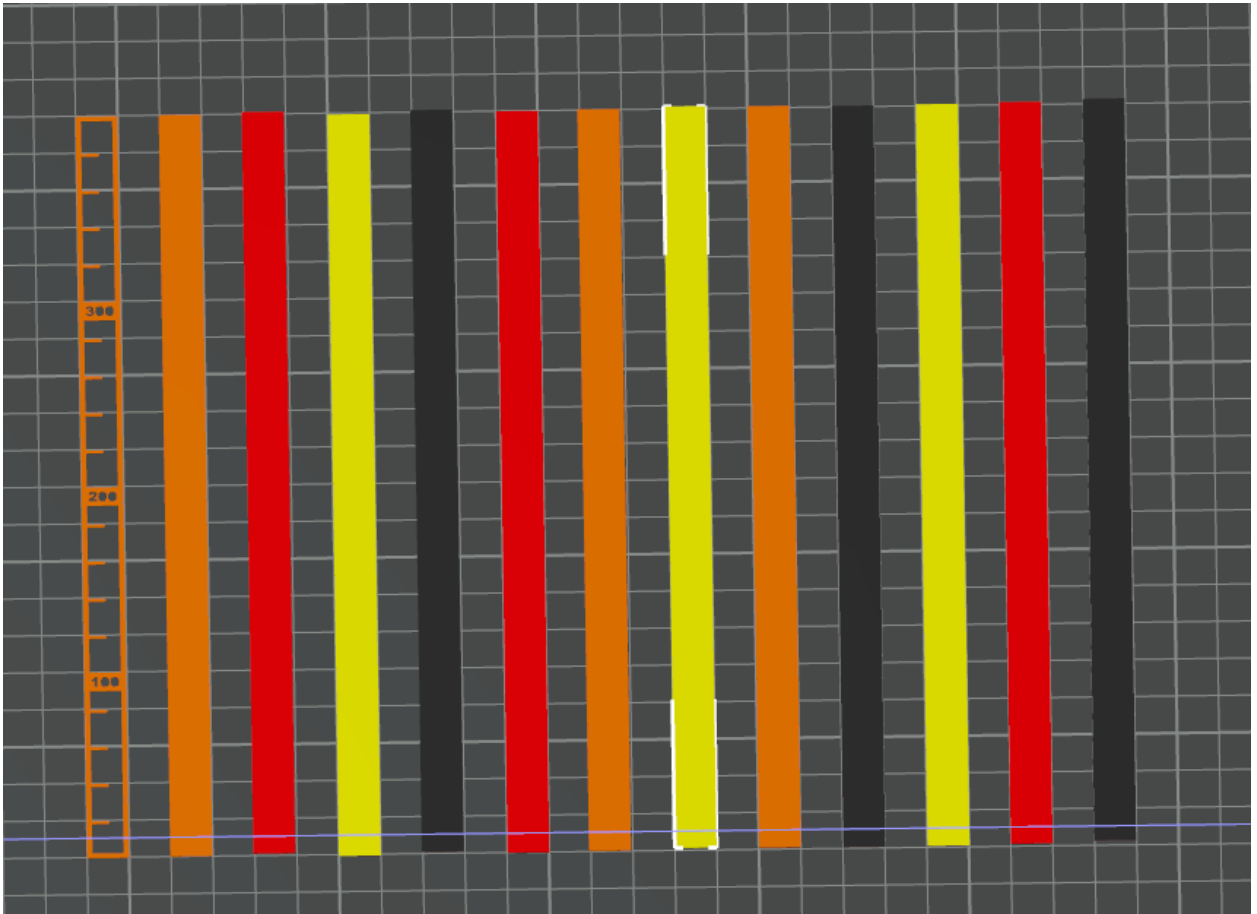
Multiplier

The multiplier should be in range [0, 3].

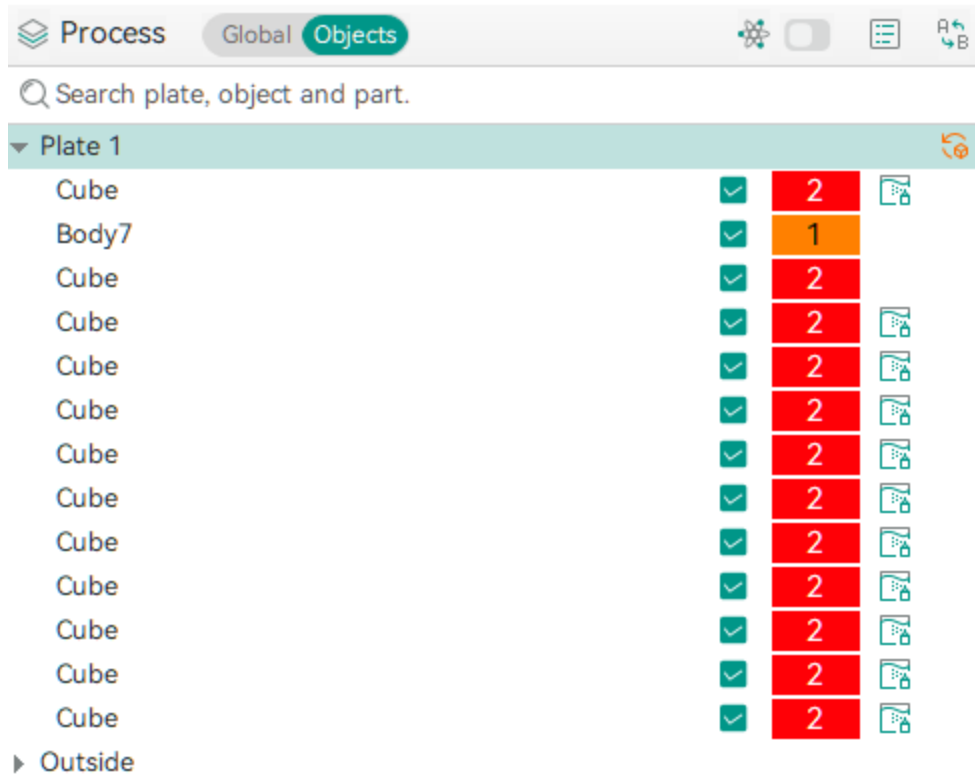
OK Cancel

7. To determine the swaps needed, you can start in column 1 and go down the column. Then go to the next column and repeat. Just be aware that some of them will repeat if you are not careful. I'm including the order I did then in.
8. The swaps needed for 4 lanes consist of (using the colors in my example):
 - a. Start with Orange (1) at the top of column 1 and transition to Red (2)
 - b. Red (2) is loaded, so transition to Yellow (3)
 - c. Yellow (3) to Black
 - d. Go to column 2, and go Black (4) to Red (2)
 - e. Red (2) to Orange (1)
 - f. Orange (1) to Yellow (3)
 - g. Go to column 3 (Yellow to black is done already) Yellow (3) to Orange (1)
 - h. Orange (1) is loaded and needs to finish with Black (4)
 - i. Black(4) transition to Yellow (3)
 - j. Yellow (3) to Red (2)
 - k. Red (2) to Black (4)
 - l. Black (4) to Orange (1)

9. That is only 11 swaps, but there are 12 swatches of filament as shown in the screenshot below. In my screenshot below, I inadvertently left out the Orange (1) to Black (4) swap. So, set the ruler on the far left to Black (4) and then it will swap properly.



10. The last setting needed is to go to the Others tab and change the print order to “By Object”. Honestly, it makes no difference for this print, since it’s a single layer, but it does force the slicer to do the objects in the sequence specific in the Objects process view. You can verify the order in which the swatches will print by selecting them individually in the Objects view.



11. If you select each object, Orca slicer will highlight it on the build plate. From Left to right, make sure that the transitions match the order you have them – i.e. when you slice the plate, you can move the vertical bar on the right and watch the order the swatches will print in. If the order is incorrect, rearrange the objects in the list of objects until, when sliced, the print order is correct.

12. When it is set up as desired, it's time to print and profit.

Date	Changed by	Changes made
6 Apr 2026	Joel Hopper	Initial Draft
9 Apr 2026	Joel Hopper	Fixed issue with first to last filament swap